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Syngas Compressor High Vibration Diagnosis and Resolution

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Abstract

This case study is focused on accurately identifying the root cause of the gradual vibration rise and intermittent vibration excursions leading to potential unavailability of a compressor train.

The subject machine train is a Synthesis gas compressor used in a fertilizer plant in middle east, and it consists of a LP and HP compressors driven by a Steam turbine in a single shaft configuration. This is a high critical compressor delivering 215,800 kg/hr of syngas at a rated speed of 9370 rpm, suction and discharge conditions are 30.4 bar(g), 40°C and 200 bar(g), 40°C respectively.

The machine was in continuous operation since last start up in April-2022 following a major overhaul. LP Compressor NDE bearing vibration trend plots showed rising trend since late 2022, mainly due to synchronous 1X vibration gradual increase resulted from suspected rubbing.

LP Compressor bearings vibration trend plots also showed multiple intermittent vibration excursion events and the amplitude during one of vibration excursion events crossed OEM Alarm limit at NDE bearing with symptoms of potential flow turbulence. LP Compressor DE vibration trends also showed similar symptoms with less amplitudes.

Based on the vibration data diagnosis it was concluded that change in rotor dynamic stiffness condition caused the change/increase in 1X amplitudes whereas the flow turbulence at particular operating conditions was driven by suspected changes in seals clearances. Thus, the decision was taken to shut down the machine train to inspect LP compressor rotor internals and inspect the compressor seals condition (interstage, center labyrinth, dry gas, and balance drum seals) and clearances for signs of deterioration. The inspection revealed significant damage to the balance drum seal.

Introduction

Machine details

The syngas compression train consists of three major casings — LP compressor (Model # 2BCL509), Steam Turbine driver (TK-431), and HP compressor (Model # 2BCL459A) — arranged in a single-shaft configuration with steam turbine driving LP and HP Compressors through flexible couplings. The OEM for the entire train is Nuovo Pignone.

The machine train is designed for continuous operation at a nominal speed of 9370 rpm, compressing synthesis gas for ammonia synthesis application. Both LP and HP Compressors are of multi-stage horizontally split barrel type, and they have tandem dry gas seals with nitrogen buffer gas at Drive-End (DE) and Non-Drive-End (NDE) bearings. The first and second critical speeds of LP Compressor rotor are 3600 rpm and 12000 rpm, respectively.

Each casing is supported on two tilting pad radial bearings (DE and NDE). All the radial bearings are equipped with two orthogonal proximity probes mounted at X (45°R) and Y (45°L) locations to monitor radial vibration, and the rotation direction is Clockwise viewed from HP compressor towards LP Compressor. Additionally, each casing contains a thrust bearing, monitored by two axial thrust position probes. A single Keyphasor probe is mounted on the shaft, providing speed and phase angle reference for the entire train.

The machine is protected by Bently Nevada 3500 rack vibration monitoring system. Radial bearings vibration Alarm and Danger limits are 85 µm pk-pk and 129 µm pk-pk, respectively. Thrust position monitoring is also configured with Alert and Danger limits as follows:

Steam Turbine -0.82 to 0.82 mm for Alert and -0.9 to 0.9 mm for Danger, LP and HP Compressors -0.5 to 0.5 mm for Alert and -0.7 to 0.7 mm for Danger.

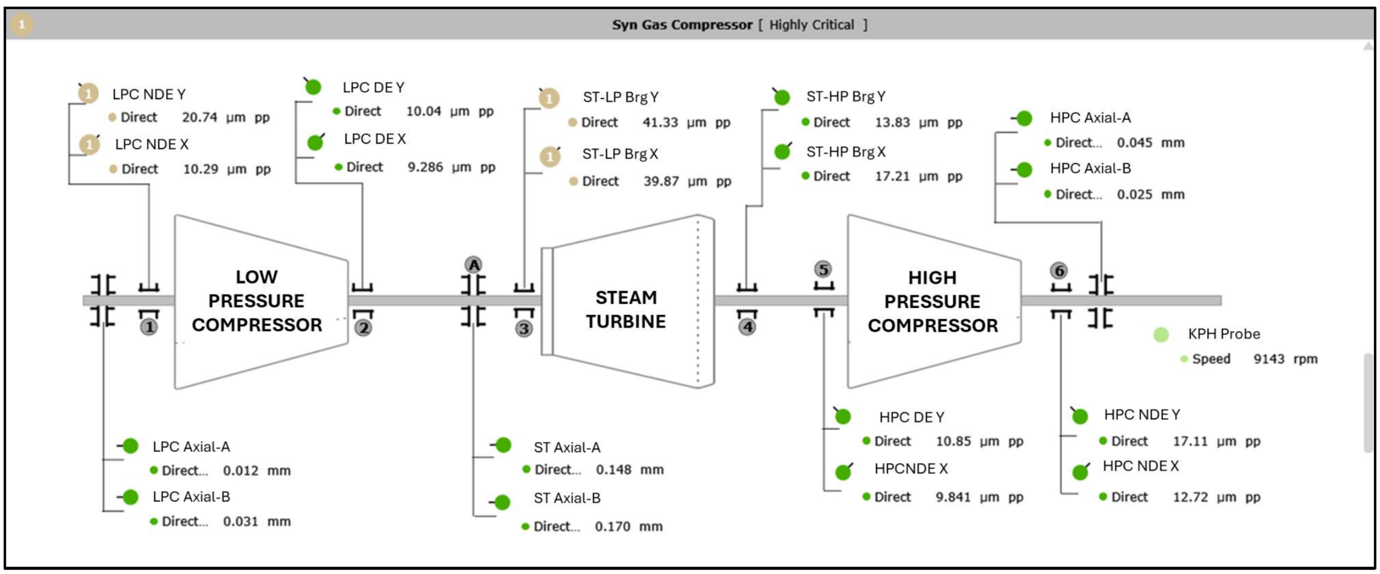

Figure 1—Machine Train Diagram showing the Steam Turbine driving Low Pressure and High-Pressure compressors coupled on either side of the steam turbine. The machine train also shows the radial vibration orthogonal proximity probes, axial probes monitoring the thrust collar position and Keypahsor probe which is used for machine train speed measurement and 1X per rev reference

Problem Statement

The syngas compressor train has been operating continuously since its last startup in April 2022. A gradual increase in radial vibration was observed at the LP Compressor NDE bearing from late 2022. In addition, intermittent vibration excursions were noted at both LP Compressor NDE and DE bearings. Notably, the amplitude of these intermittent vibration excursions was high on at the NDE bearing compared to the DE. The severity of the vibration amplitudes during the intermittent excursions increased with time and the amplitudes reached close to hardware Alarm setpoint in June-2023 on LP Compressor NDE bearing. In contrast, the radial vibration levels at the HP compressor and steam turbine bearings remained stable, with no similar intermittent vibration excursion events observed. Furthermore, LP, HP, and steam turbine thrust bearings axial position values exhibited stable trends. This localized vibration behaviour at the LP compressor NDE bearing prompted further investigation.

Vibration Data Diagnosis and Data Presentation

The machine train was in continuous operation since April-2022 with minor changes in speed as per the plant production and operational requirement, System 1 plot below shows the speed trend in the year 2022-23, please see Fig. 2.

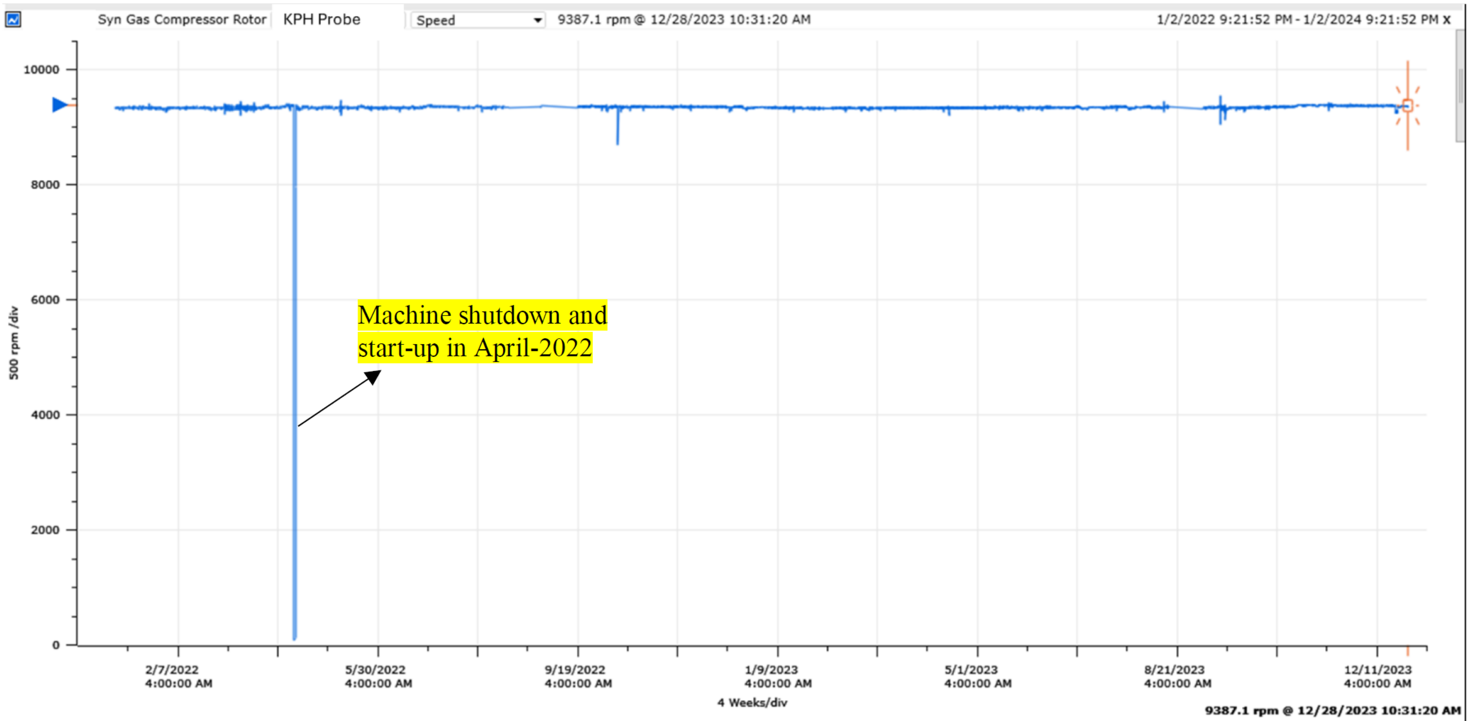

Figure 2—Machine speed trend in the year 2022-23. The last shutdown and start-up of the machine was in April-2022 and the machine has been in continuous operation since then indicating the criticality of the compressor in plant's production.

The steam turbine and HP Compressor radial bearings showed stable vibration trends with dominant 1X amplitudes during the machine operation in 2022-23. LP Compressor DE and NDE bearings also showed below Alarm limit and dominant 1X amplitudes during the machine operation in 2022. However, the vibration trends started showing abnormal vibration behavior since the last start-up in April-2022. Table 1 below shows the vibration amplitudes and thrust position values across all bearings of the machine train during steady state operation after the start-up in April-2022.

Table 1—Vibration amplitudes and thrust position values across all bearings of the machine train during steady state operation after the start-up in April-2022. The vibration amplitudes and thrust position values were inside the alarm boundary limits in April-2022 prior to the start of the abnormal vibration behavior at LP Compressor NDE bearing

S. No.	Measurement Location	Direct Amplitude	1X Amplitude	1X Phase Angle	Alarm limit	Danger limit
1	LP Compressor DE Bearing - Y-probe	10.1 µm pp	6.8 µm pp	195 °	85 µm pp	129 µm pp
2	LP Compressor DE Bearing - X-probe	9.2 µm pp	3.9 µm pp	201 °	85 µm pp	129 µm pp
3	LP Compressor NDE Bearing - Y-probe	20.7 µm pp	16.4 µm pp	36 °	85 µm pp	129 µm pp
4	LP Compressor NDE Bearing - X-probe	10.3 µm pp	3.1 µm pp	128 °	85 µm pp	129 µm pp
5	Steam Turbine NDE Bearing - Y-probe	41.3 µm pp	34. 7 µm pp	232 °	85 µm pp	129 µm pp
6	Steam Turbine NDE Bearing - X-probe	39.8 µm pp	34.6 µm pp	342 °	85 µm pp	129 µm pp
7	Steam Turbine DE Bearing - Y-probe	13.8 µm pp	7.26 µm pp	187 °	85 µm pp	129 µm pp
8	Steam Turbine DE Bearing - X-probe	17.2 µm pp	12.7 µm pp	316 °	85 µm pp	129 µm pp
9	HP Compressor DE Bearing - Y-probe	10.8 µm pp	8.2 µm pp	186 °	85 µm pp	129 µm pp
10	HP Compressor DE Bearing - X-probe	9.8 µm pp	6.9 µm pp	279 °	85 µm pp	129 µm pp
11	HP Compressor NDE Bearing - Y-probe	17.1 µm pp	13.2 µm pp	123 °	85 µm pp	129 µm pp
12	HP Compressor NDE Bearing - X-probe	12.7 µm pp	8.8 µm pp	216 °	85 µm pp	129 µm pp
	Measurement Location	**Thrust position value**			**Alarm limit**	**Danger limit**
13	LP Compressor Thrust position-A	11.7 µm			-500 to 500 µm	-700 to 700 µm
14	LP Compressor Thrust position-B	31.1 µm			-500 to 500 µm	-700 to 700 µm
15	Steam Turbine Thrust position-A	148.2 µm			-820 to 820 µm	-900 to 900 µm
16	Steam Turbine Thrust position-B	169.6 µm			-820 to 820 µm	-900 to 900 µm
17	HP Compressor Thrust position-A	45.2 µm			-500 to 500 µm	-700 to 700 µm
18	HP Compressor Thrust position-B	24.5 µm			-500 to 500 µm	-700 to 700 µm

The LP compressor NDE bearing vibration levels were relatively low, with amplitudes less than 25 µm pk-pk during early 2022. However, over time, a gradual increasing trend in Direct and 1X vibration amplitudes from early 2022 to late 2023 was observed, particularly accelerating from mid-2023 onward. Superimposed on this increasing trend were several intermittent vibration excursions, beginning as early as late 2022 and becoming more frequent and severe through 2023. These transient events were observed in both X and Y probes, with multiple excursions reaching close to Alarm limit in the 3500 rack in June-2023. Toward the end of December 2023, the overall vibration amplitudes reached up to 55 µm pk-pk with dominant 1X behavior, while the Alarm limit is at 85 µm pk-pk. Please see Fig. 3 below showing this abnormal behavior.

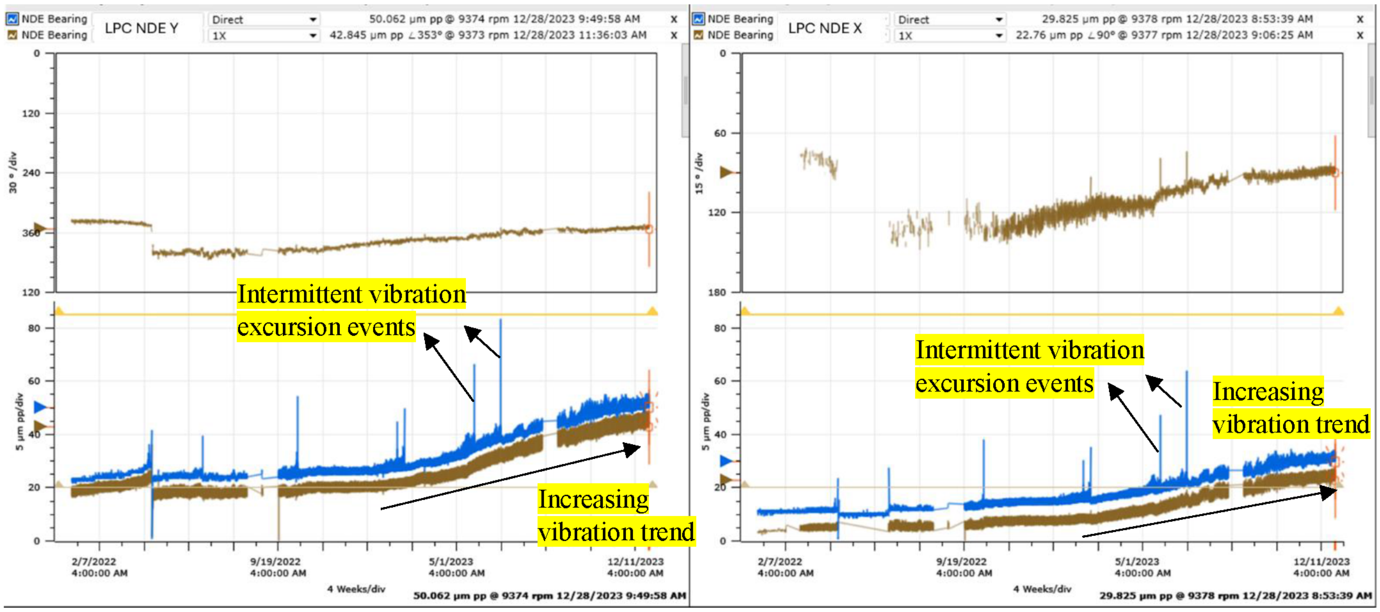

Figure 3—LP Compressor NDE bearing vibration trend plots – 2022-23. The plots showed increasing vibration trend and there were multiple intermittent vibration excursion events. Also, the vibration behavior was 1X dominant and the 1X vector stayed stable when there were intermittent vibration excursion events on Direct vibration trends.

The LP compressor DE bearing vibration levels remained relatively stable during early 2022, with Direct vibration amplitudes below 20 µm pk-pk. However, starting from late 2022, intermittent vibration excursions events began to emerge, though lower magnitude compared to the NDE bearing. The vibration levels during these intermittent excursions reached a maximum amplitude of ~55 µm pk-pk in June-2023 which is below the hardware alarm limit of 85 µm pk-pk. DE bearing Direct and 1X vibration amplitudes started to show notable fluctuations towards end of December-2023, however, the amplitudes stayed well below Alarm limit (< 20 µm pk-pk), as shown in Fig. 4.

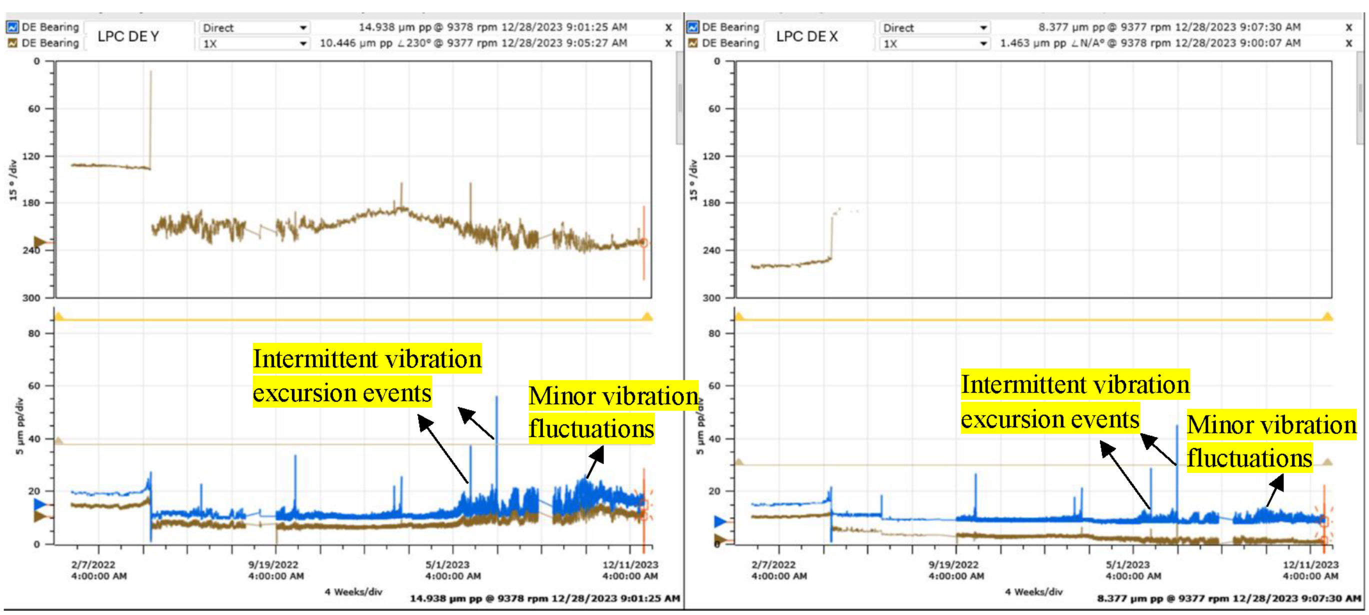

Figure 4—LP Compressor DE bearing vibration trend plots – 2022-23. The plots showed multiple intermittent vibration excursion events, however, the vibration trends did not show gradual rise behavior similar to what was seen at NDE bearing. Also, 1X vector stayed stable when there were intermittent vibration excursion events on Direct vibration trends.

Review of LP compressor NDE (blue and green traces) and DE (gray and red traces) bearings overlay Direct vibration trend plots showed simultaneous vibration excursions on all the probes as can be seen in Fig. 5, indicating that the anomalous vibration behavior is not instrumentation-related but rather driven by a genuine mechanical issue within the machine. It is important to note that, unlike the NDE bearing which exhibited a gradual and continuous increase in vibration amplitude over time, the DE bearing did not show such a trend. Instead, the DE bearing displayed low-magnitude vibration fluctuations, particularly noticeable during the second half of 2023. The higher vibration severity at the NDE bearing, compared to the DE, further implies that the vibration response is higher at the NDE bearing location.

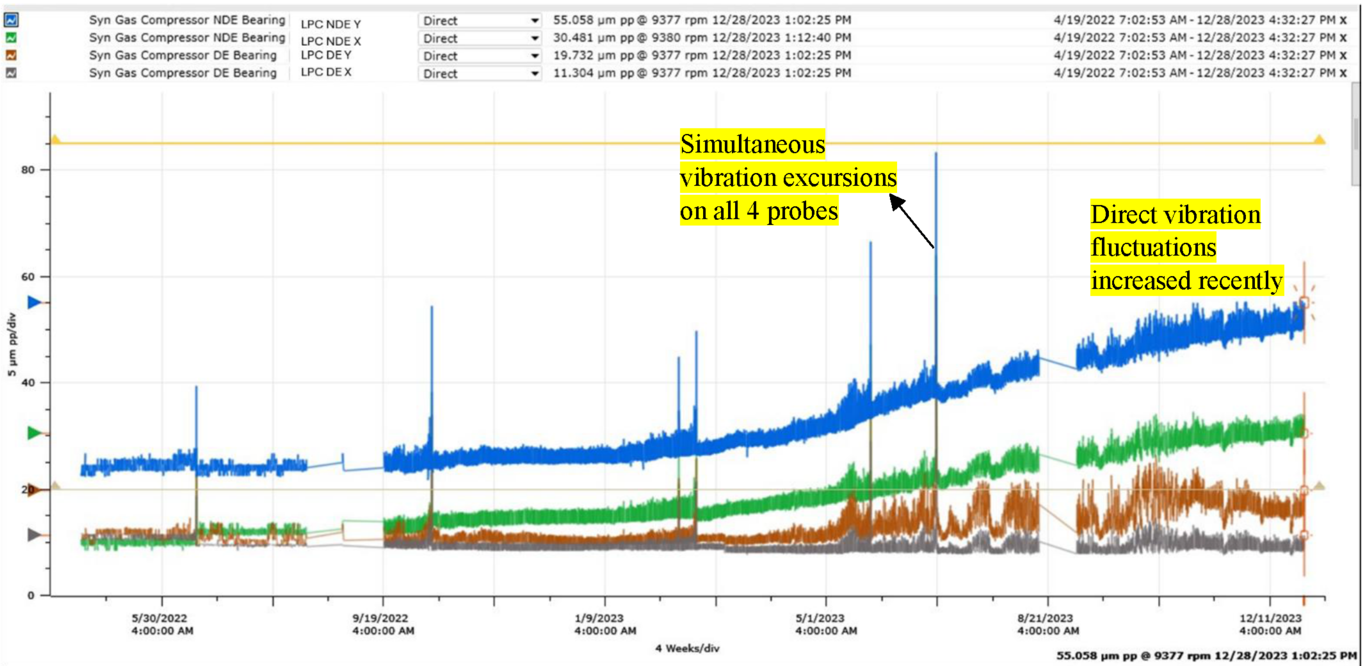

Figure 5—LP Compressor NDE and DE bearings Direct vibration trend plots – 2022-23. The trend plots showed simultaneous vibration excursions at both the bearings probes and the Direct vibration fluctuations have increased on DE bearing recently.

The vibration data plots for the LP compressor NDE bearing from System 1 below (Fig. 6) between Oct-2022 and Dec-2023 showed gradual rise in X and Y-probes amplitudes. The corresponding full-spectrum asynchronous waterfall plot highlighted a dominant and increasing 1X component. The slow-roll compensated shaft centerline plot revealed no significant movement in the rotor's position during the vibration increase period. However, the overlay of slow-roll compensated direct orbit plots between the low-vibration (Oct 2022) and high-vibration (Dec 2023) periods showed a marked increase in orbit shape along with a distinct constraint movement (flattened orbit at one-side) indicating a unidirectional force (also known as preload) acting on the rotor. Overall, the vibration plots indicate that the NDE bearing is subject to a developing mechanical anomaly characterized by increasing 1X vibration amplitude and preload symptom.

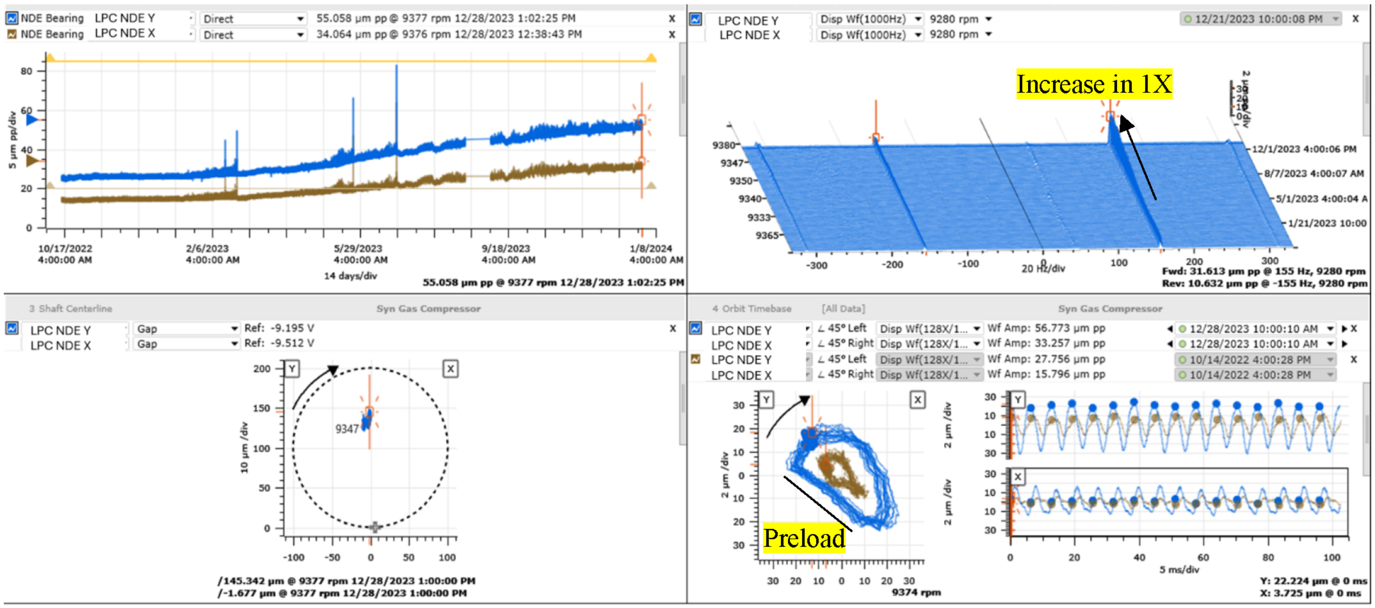

Figure 6—LP Compressor NDE bearing probes Direct vibration, full spectrum waterfall, slow roll compensated shaft centerline plot and overlay Direct orbit plots between Oct-2022 (low vibration) and Dec-2023 (high vibration). Waterfall plot showed increase in 1X amplitude, SCL plot showed minimal change in position during the vibration increase and Direct orbit plot showed increase in amplitude with preload symptom.

The vibration data plots for the LP compressor DE bearing from System 1 below (Fig. 7) between Oct-2022 and Dec-2023 showed minor Direct vibration fluctuations since mid-2023. The corresponding full-spectrum asynchronous waterfall plot showed a minor rise in 1X component along with minor sub-synchronous vibration component. The slow-roll compensated shaft centerline plot revealed no significant movement in the rotor's position similar to what was seen in NDE bearing. The overlay of slow-roll compensated Direct orbit plots showed minor unstable orbit in Dec-2023 compared to Oct-2022, however, it is to be noted that the overall amplitudes were low, less than 20 µm pk-pk.

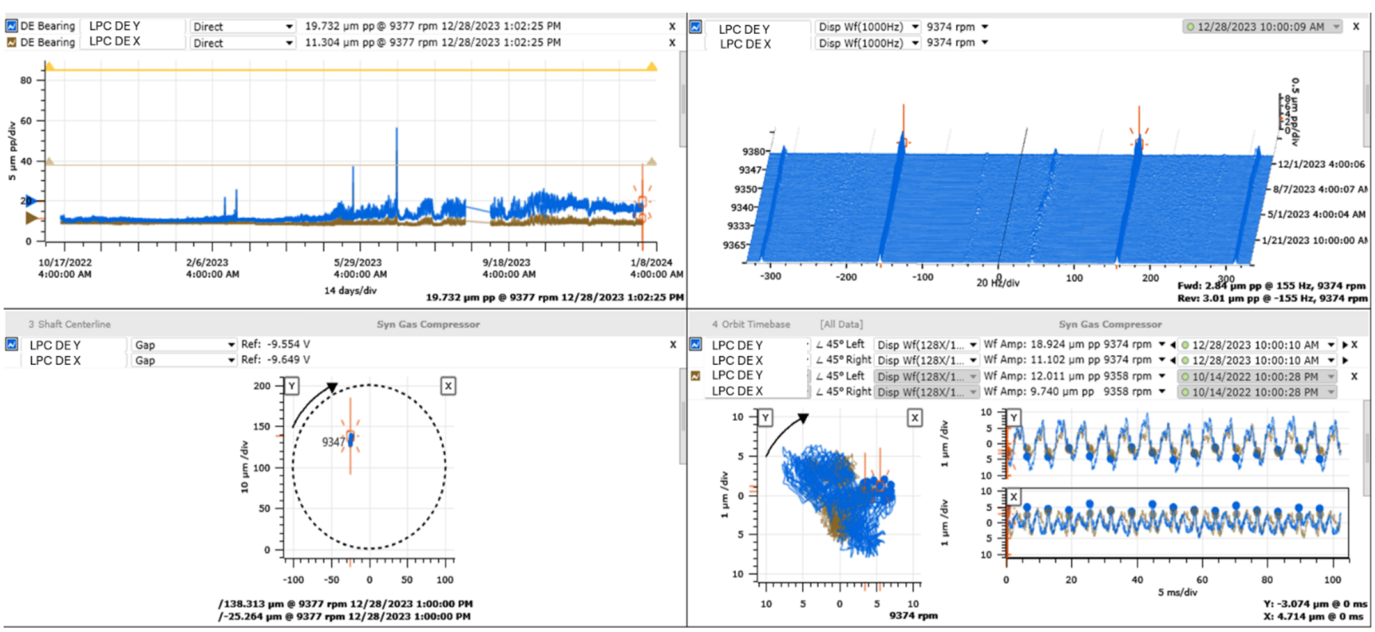

Figure 7—LP Compressor DE bearing probes Direct vibration, full spectrum waterfall, slow roll compensated shaft centerline plot and overlay Direct orbit plots between Oct-2022 (stable vibration) and Dec-2023 (minor vibration fluctuations).

The below vibration plots show the LP compressor NDE bearing vibration behavior during one of the high vibration excursion events which happened on 25-June-2023, please see Fig. 8. LP Compressor NDE bearing Y-probe exhibited minor vibration fluctuations leading up to a sharp excursion, followed by a return to lower, stable levels, similar behavior was seen on X-probe as well as described in the previous vibration plots. The asynchronous waterfall plot and overlay spectrum revealed a distinct sub-synchronous vibration component at ~0.3X, which was more pronounced during the fluctuation and high vibration event and significantly reduced afterward. Additionally, the overlay of slow-roll compensated Direct orbit plots showed a minor instability behavior near the high vibration event. Collectively, these observations suggest process fluid turbulence issue, driven by sub-synchronous excitation.

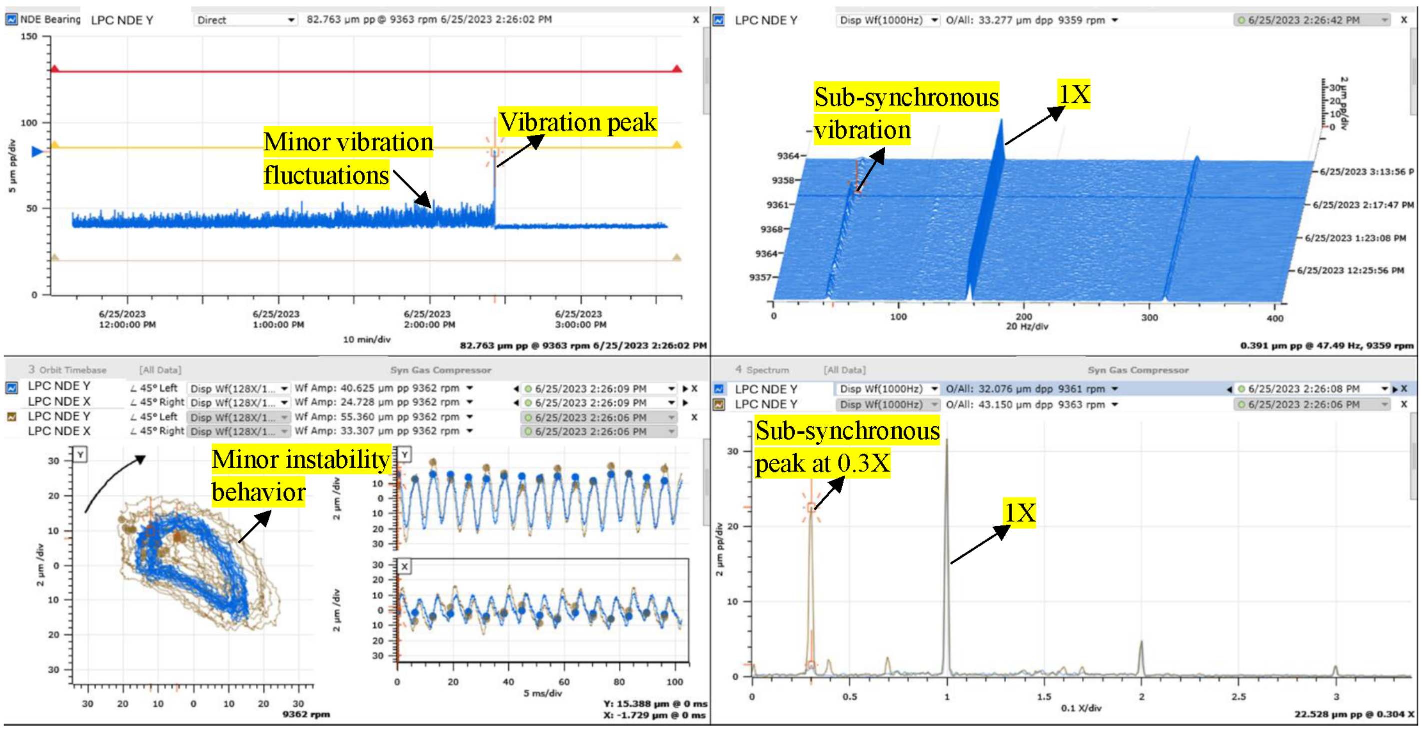

Figure 8—LP Compressor NDE bearing vibration trend zoom view during the recent vibration excursion event on 25-Jun-23. Review of overlay spectrum plots between high vibration event and after the high vibration event showed sub-synchronous vibration peak at 0.3X during the high vibration event and review of overlay Direct orbit plots showed fluid turbulence behavior during the high vibration event.

Review of another vibration excursion event at LP Compressor NDE bearing (on 23-Mar-23) also showed similar vibration behavior, i.e., sub-synchronous vibration and process fluid turbulence behavior (please see Figure 13), however, the amplitudes were lower compared to the event on 25-Jun-23.

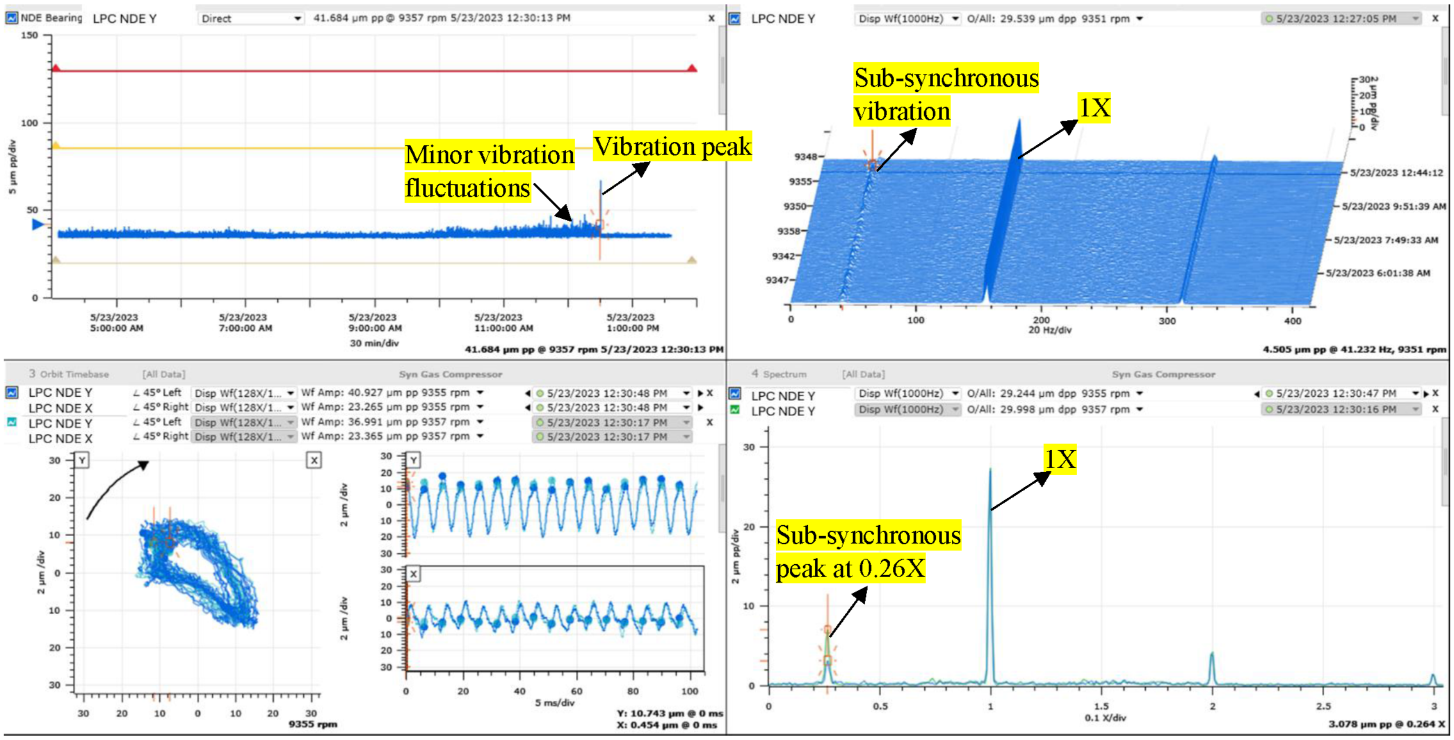

Figure 9—LP Compressor NDE bearing vibration trend zoom view during the vibration excursion event on 23-Mar-23. Review of overlay spectrum plots between high vibration event and after the high vibration event showed sub-synchronous vibration peak at 0.26X during the high vibration event.

System 1 does not have bearing temperatures data to review and correlate with the vibration changes. Thus, the bearings health during the vibration changes was unknown. Also, site does not have DCS data during the events as they do not store data for more than 3 months. Thus, the vibration changes could not be correlated with the process changes during the sub-synchronous vibration events and during the gradual rise in vibration.

Note: Site invited Baker Hughes Cordant™ to the site for diagnosis when the vibration levels were reaching the hardware Alarm limit. Thus, the bearing temperatures and process data during the high vibration events was not available as the last high vibration excursion event was more than 3 months old.

Recommendations and Action Plan

1. Based on the comprehensive diagnosis of System 1 vibration data, the following primary contributing factors were suspected for the abnormal LP Compressor vibration behaviour:

    a. A change in rotor balance condition (likely due to material loss, or deposit buildup) and/or dynamic stiffness condition caused the change/increase in 1X amplitudes.
    b. Process fluid-induced turbulence, particularly sub-synchronous excitation events (notably at ~0.3X), possibly linked to changes in interstage labyrinth seal, dry gas seal, or balance drum seal clearances. The increased seal clearances can reduce damping and increased cross-coupled stiffness, thereby promoting fluid-induced turbulence vibration excitation.
2. As the LP Compressor NDE bearing vibration approached alarm limits with clear symptoms of preload and process fluid-induced turbulence, the following inspection and action plan was agreed upon with the customer:

    a. Disassemble and inspect the LP compressor rotor and internal elements for signs of mass imbalance (e.g., erosion, fouling/deposits) and sources of preload (suspected rubbing).

b. Inspect and measure the condition and clearance of all internal sealing elements, including:

   i. Interstage labyrinth seals
   ii. Centre labyrinth seal
   iii. Dry gas seals
   iv. Balance drum seals

Findings during inspection

Based on Bently Nevada System 1 data review and recommendations, customer decided to shut down the compressor for major inspection and following are the findings:

1. The inspection confirmed significant wear and damage to the LP compressor balance drum labyrinth seal (please see Fig. 10) thus the potential cause for the intermittent vibration fluctuations is deemed to be process gas leakage through the seals causing process fluid-induced turbulence vibration excitation.
2. The inspection also revealed significant erosion at the thrust collar (please see Fig. 11). It is to be noted that the thrust collar is near NDE bearing thus the potential cause for the gradual rise in vibration is deemed to be the erosion (mass loss) from the thrust collar.

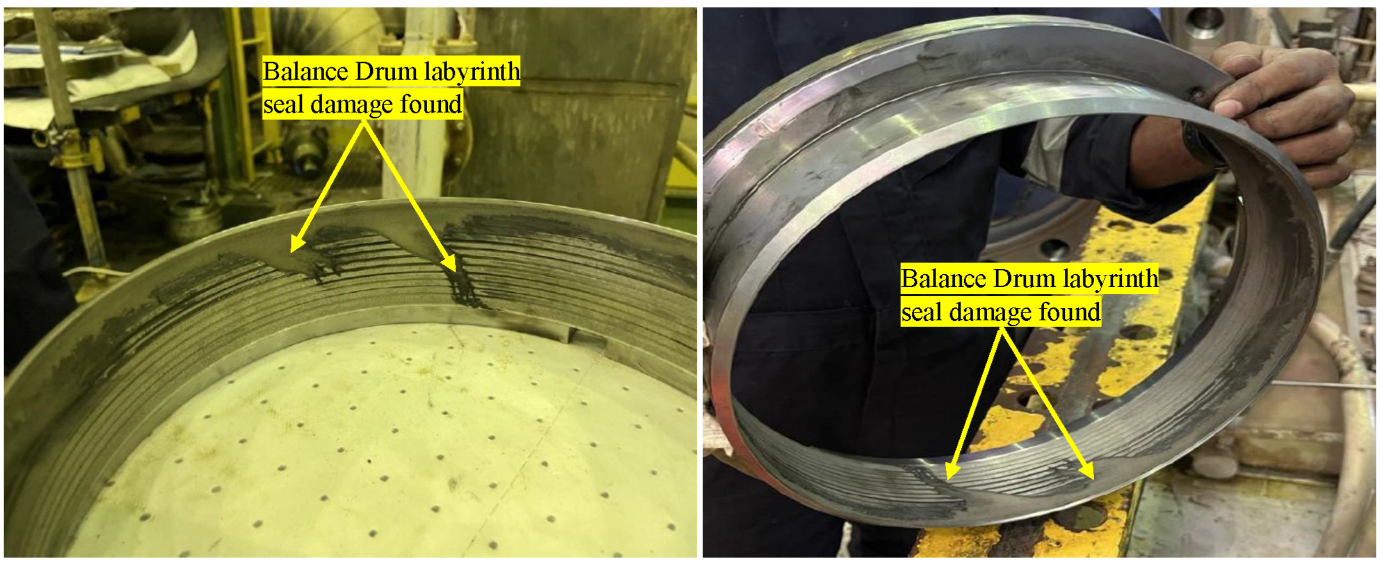

Figure 10—LP Compressor balance drum seals as found damage pictures during the inspection. Photographs show significant rubbing of the labyrinth seals around the circumference and loss of seal material.

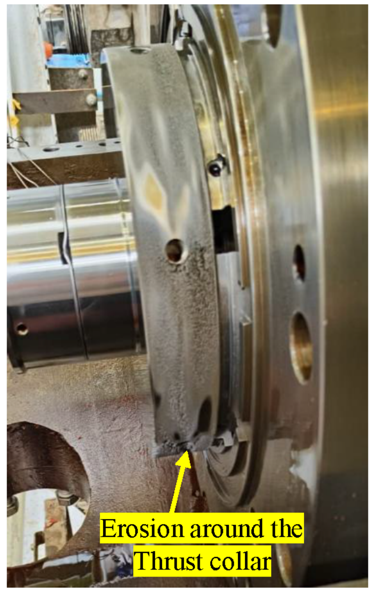

Figure 11—LP Compressor Thrust collar erosion found during the inspection leading to mass loss from the thrust collar

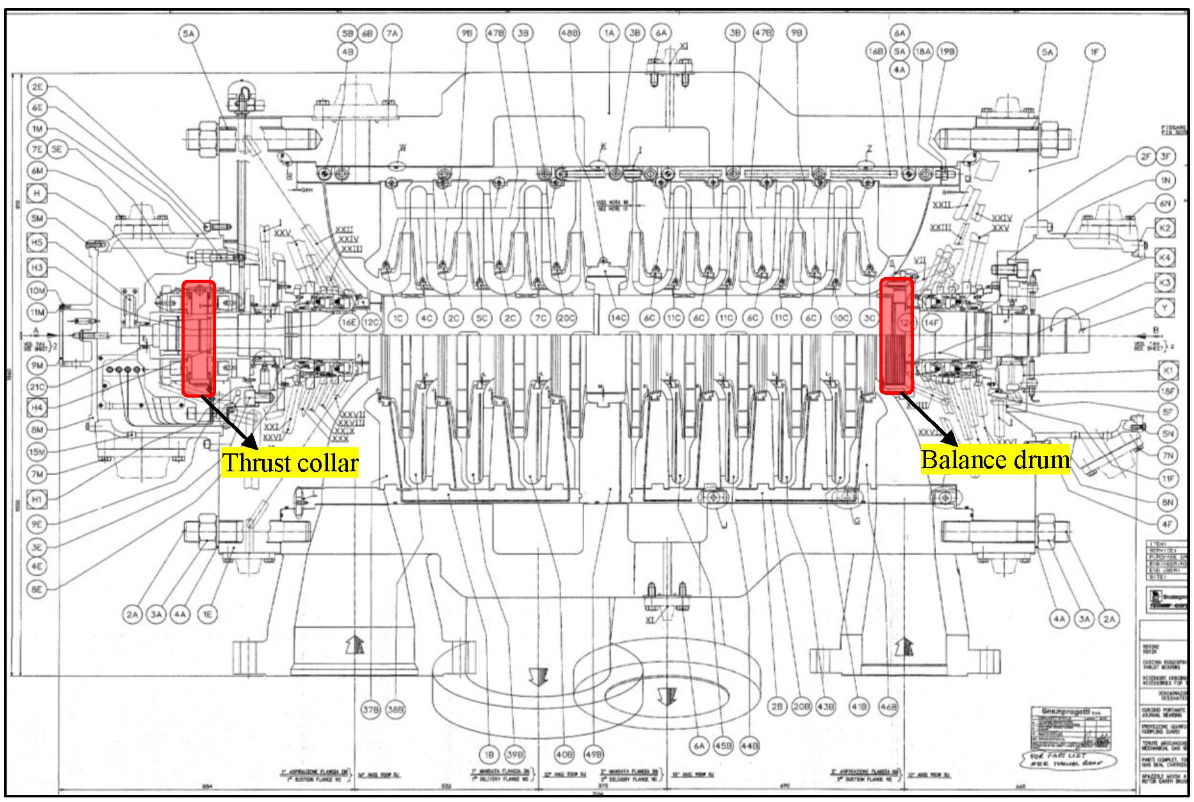

Figure 12—LP Compressor Cross section drawing showing thrust collar and Balance drum locations. The thrust collar is located at the NDE bearing of the compressor and the balance drum is located at DE bearing of the compressor.

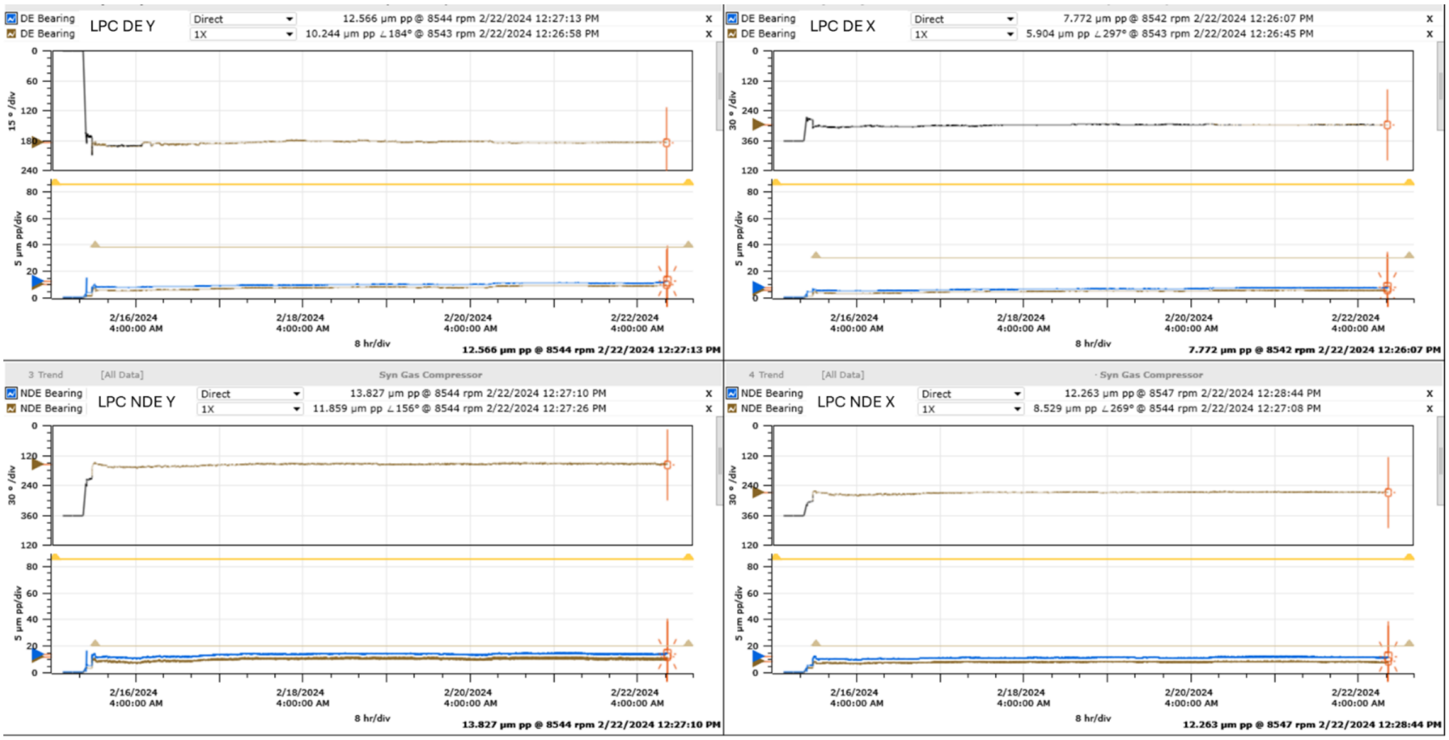

Figure 13—LP Compressor DE and NDE bearings vibration trend plots after the rotor replacement. The vibration levels stayed well below Alarm limit and there were no intermittent vibration excursions or sub-synchronous vibration behavior was seen following the overhaul.

Conclusion: Probable Failure Mechanism Summary

The LP syngas compressor exhibited increasing 1X vibration amplitudes and multiple sub-synchronous vibration excursions, primarily at the NDE bearing. Detailed vibration diagnosis of System 1 data pointed to both synchronous and fluid-induced turbulence mechanisms. Upon shutdown and inspection, two major mechanical issues were identified:

1. **Severe damage to the balance drum seal:** This caused elevated leakage of high-pressure gas across the seal, reducing axial balancing effectiveness. The resulting cross-coupled fluid forces likely triggered sub-synchronous whirl, seen as 0.3X vibration during multiple vibration excursion events.
2. **Erosion of the thrust collar near the NDE bearing:** This introduced rotor mass imbalance consistent with the observed gradual 1X vibration increase at the NDE bearing.

Although the mechanical degradation was physically located near both the DE and NDE ends, the vibration response was high at the NDE bearing side which could be attributed to relatively lower damping or dynamic stiffness, or higher modal sensitivity at the operation speed at the NDE bearing. These findings confirm that fluid-induced excitation from the degraded balance drum and mechanical imbalance from thrust collar erosion were the combined root causes of the observed vibration behavior.

Machinery Vibration Behavior Post Corrective Actions

Customer replaced the LP Compressor rotor following the inspection findings. The compressor was back in normal operation after rotor replacement and the vibration levels stayed well below alarm limit (max vibration < 15 μm pk-pk) and there were no intermittent vibration excursions seen indicating that the issue was successfully resolved, please see Fig. 13.

Lessons learnt and Enhancements proposed

1. Early subtle vibration excursions can be easily overlooked: The early presence of subtle vibration fluctuations and excursions, particularly those associated with fluid-induced turbulence, can often be mistaken for spurious instrumentation noise. This case highlighted the importance of conducting a detailed review of both static trends and waveform data during early-stage anomalies.

   Enhancement proposed: Optimize the condition monitoring alarms and configure spectral bands (e.g., sub-synchronous bands) in System 1 to enable automated detection and trending of such subtle changes.
2. Lack of integrated process and bearing temperature data limited diagnostic correlation: During this case, the absence of archived DCS process parameters and bearing metal temperature trends restricted the ability to correlate vibration anomalies with changes in process conditions or bearing health.

   Enhancement proposed: Integrate compressor process parameters (pressure, temperature, flow, valves position changes etc.) and bearing metal temperatures into System 1. This would allow condition monitoring engineers to correlate the vibration changes with the machine train process changes and to assess bearings healthiness.

Overall, this case study presents a comprehensive diagnostic analysis of high radial vibration and intermittent excursions observed at the LP Compressor NDE bearing of a syngas compressor train. Through in-depth data review using Bently Nevada System 1, the investigation identified rotor dynamic stiffness changes, preload, and seal degradation—particularly balance drum seal damage and thrust collar erosion—as primary contributors to increasing 1X and sub-synchronous vibration components.

Key findings include:

1. Gradual increase in vibration amplitudes post-April 2022 restart.
2. Dominant 1X and sub-synchronous (~0.3X) excitation associated with reduced rotor damping.
3. Hardware inspection confirmed mass loss at the thrust collar and labyrinth seal damage, validating the vibration-based diagnosis.

Lessons learned emphasize the importance of early detection of subtle vibration excursions, better correlation of process and vibration data, and refinement of alarm strategies in condition monitoring systems.

Future enhancements proposed include:

1. Optimizing System 1 alarms and spectral bands.
2. Integrating compressor process parameters and temperature trends with vibration analysis.
3. Performing structured rotor and seal inspections during major outages to prevent recurrence.

The findings not only resolved a critical reliability issue for the plant but also highlight best practices in condition monitoring and root cause diagnosis, contributing valuable knowledge to the field of turbomachinery diagnostics.

Nomenclature

ST	Steam Turbine
HP	High Pressure
LP	Low Pressure
DCS	Distributed Control System
NDE	Non-Drive End
DE	Drive End

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